

JEE MAIN REVISION MASTER: SEMICONDUCTOR DEVICES AND LOGIC GATES 2026 3 QUESTIONS

5% WEIGHTAGE $\uparrow$ 50.15%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 32 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Semiconductor Devices And Logic Gates - Core Application	32 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Semiconductor Devices And Logic Gates - Core Application (Very High Yield — 32 PYQs)

Core Principle: Key problem pattern in Kcet Semiconductor Devices And Logic Gates.

■■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A wafer of pure germanium crystal has two parts X and Y. The end X is obtained by doping with arsenic and Y with indium. It is connected to ..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In which of the following figures, diode is reverse biased?..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "An n-type and p-type semiconductor can be obtained by respectively doping pure silicon with..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A wafer of pure germanium crystal has two parts X and Y. The end X is obtained by doping with arsenic and Y with indium. It is connected to a battery as shown in the figure. Which of the following statements is correct?

- (A) X is p-type, Y is n-type and the junction is forward biased
- (B) X is n-type, Y is p-type and the junction is forward biased
- (C) X is p-type, Y is n-type and the junction is reverse biased
- (D) X is n-type, Y is p-type and the junction is reverse biased

Correct Answer: None

Solution Analysis: Part X is doped with arsenic. Arsenic is a Group 15 element, so it has 5 valence electrons. When it is added to germanium, it gives extra free electrons. So, part X becomes an n-type semiconductor. Part Y is doped with indium. Indium is a Group 13 element, so it has 3 valence electrons. When it is added to germanium, it creates holes. So, part Y becomes a p-type semiconductor. In the battery symbol, the longer line represents the positive terminal, and the shorter line represents the negative terminal. From the figure, part X is connected to the positive terminal and part Y is connected to the